



Learning Python - Sheet #2

Functions – If statements

1. Functions

A function is a way to define a piece of code that can be reused. It can take some parameters.

Try this:

```
def greetings(name):  
    print("Hello %s" % name)  
  
greetings("Richard")  
greetings("Elizabeth")
```

Notice how IDLE shows “...” instead of “>>>” when it’s waiting for you to type the function content (which is called the **function body**) and how it automatically adds some spaces before the lines (called **indentation**).

Note also that we used **print** to display the message. A function can also return something. Our **greetings** function doesn’t return anything, but this one does:

```
def pythonista(name):  
    return "%s is a pythonista" % name  
  
pythonista("Guido")
```

Functions are very important, so let’s look at another example. This one takes multiple parameters and returns something too.

```
def mix(a, b, c):  
    return a*b + b*c + c*a  
  
mix(1, 2, 3)
```

2. If statements

Sometimes, you want your code to do different things depending on some condition. One way to do this is with if statements. To make it easier to experiment, let’s put them in functions, though they don’t need to be:

```
def greet(name, age):  
    print("Hello %s" % name)  
    if age > 18:  
        print("I see you’re an adult")  
  
greet("Mariet", 21)  
greet("Hugo", 9)
```



Find the bug:

You might just have written your first bug! A bug is an error in a program. Can you find it?

You can also add an else after an if:

```
def tell_secret(password):  
    if password == "abc123":  
        print("The secret is that ")  
    else:  
        print("Sorry, I can’t tell you the secret")
```



Have fun experimenting:

You can also use **elif** if you want to do something when the condition of the **if** is not met but another is. Can you figure out how to use it?